

NGUYEN BA LONG

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Summary

Third-year Computer Science student with a solid foundational knowledge and hands-on experience through various course projects in Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and Generative AI (NLP/LLMs). Familiar with data preprocessing, building neural network architectures, training models, and designing intelligent RAG/Agent workflows. Seeking an AI/ML Intern opportunity to apply academic knowledge, learn from practical challenges, and continuously improve technical skills.

Education

University of Information Technology - VNUHCM

Sep. 2023 - Feb. 2027 (Expected)

Bachelor of Science in Computer Science

Ho Chi Minh City, Vietnam

- **GPA:** 8.23/10.0
- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Mining, Big Data.

Technical Skills

Languages: Python, C/C++, SQL, JavaScript

AI/ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, CNN, Bi-LSTM

GenAI, LLMs & RAG: LangChain, LangGraph, Prompt Engineering, BM25, Vector Search, Hybrid Retrieval, Cross-Encoder Reranking

Computer Vision & Multimodal: OpenCV, PaddleOCR, Image Segmentation, SigLIP2, Qwen2-VL

Backend & Tools: FastAPI, Flask, PostgreSQL, Milvus, Elasticsearch, Docker, Git

Projects

Temporal RAG for UIT Course Video Q&A | *LangGraph, Qwen2-VL, Whisper, FastAPI*

[GitHub](#)

- Engineered a directed state graph RAG pipeline via LangGraph for multi-modal lecture video Q&A, designing agentic workflows for efficient query routing.
- Synchronized slide text (extracted via Qwen2-VL VLM) with audio transcriptions (Whisper-large-v3) to build a structured data pipeline mapping materials to timestamps.
- Implemented a hybrid retrieval engine using Reciprocal Rank Fusion (RRF) to unify dense embeddings and BM25 keywords, refined by a Cross-Encoder reranker.
- Developed asynchronous RESTful APIs using FastAPI to minimize I/O latency; applied the RAGAS framework to benchmark and validate Faithfulness and Answer Relevancy.

AIC Multimodal Video Retrieval System | *SigLIP2, Milvus, Elasticsearch, PaddleOCR, Whisper*

[GitHub](#)

- Built a multi-modal data pipeline extracting SigLIP2 visual features, PaddleOCR text, and Whisper transcripts to prepare datasets for AI model evaluation.
- Developed a hybrid search engine combining Milvus (dense vectors) and Elasticsearch (sparse keywords), fused via Weighted Reciprocal Rank Fusion (RRF).
- Optimized retrieval accuracy with a BGE-Reranker (bge-reranker-v2-m3) to ensure precise frame-to-millisecond mapping for AI Challenge payloads.

Self-Refined LLM Reward Designer (Paper Reproduction) | *Python, Flask, LLM API, Q-Learning*

[GitHub](#)

- Reproduced and adapted a research paper on using LLMs to automatically generate and iteratively self-refine reward functions for Reinforcement Learning.
- Built a lightweight 2D environment (Gridworld) and a Q-Learning agent to enable CPU-friendly experimentation and faster prototyping.
- Developed a Flask API pipeline to orchestrate the LLM workflow: prompt engineering, Python code execution, and feedback logging for the self-refinement loop.

HTS-MultiResUNet: Dual-Attention Hybrid CNN-Transformer | *TensorFlow, Keras, OpenCV*

[GitHub](#)

- Designed a hybrid CNN-Transformer architecture integrating Squeeze-and-Excitation blocks and a Multi-Head Self-Attention (MHSA) bottleneck to extract spatial features.
- Trained and evaluated the model using 5-fold cross-validation, handling data preprocessing pipelines and spatial class imbalances.
- Implemented an Edge-Aware Focal Tversky Loss function to improve boundary preservation, outperforming the baseline MultiResUNet by 5.83% in Jaccard Index.

Languages

Languages: Vietnamese (Native)

English: Professional Reading Proficiency; TOEIC Speaking & Writing 250.